ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** heredity and variation	Genetics deals with inheritance.
2	**(d)** 46	23 pairs.
3	**(b)** Colour of the eyes	Controlled by genes.
4	**(d)** garden pea	Pisum sativum.
5	**(a)** XY	One X and one Y.
6	**(c)** A scar from an accident	Gained in life.
7	**(c)** trait	Observable feature.
8	**(c)** XX	Two X.
9	**(d)** not passed to the next generation	They do not change germ cells.
10	**(a)** nucleus	Chromosomes are in the nucleus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gregor Mendel (2).	2
12	Thread-like structures in the nucleus (1); Made of DNA and protein, carrying genes (1).	1+1
13	3 : 1 (2).	2
14	Transmission of traits from parents to offspring (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Seed shape (round or wrinkled) (1); Seed colour (yellow or green) (1); Plant height (tall or dwarf) (1).	1+1+1
16	TT gives only T (1); Tt gives T and t (1); tt gives only t (1).	1+1+1
17	Genes for different traits sort separately (1); This forms new combinations (1); It is seen in the dihybrid cross (1).	1+1+1
	OR: Dominant trait appears in F1 (1); Recessive trait is hidden in F1 (1); Recessive shows only when both alleles are recessive (1).	1+1+1
18	It helps survival in changed conditions (1); It allows adaptation (1); It leads to diversity (1).	1+1+1
	OR: A chart to show crosses (1); It lists gametes of each parent (1); It shows possible offspring genotypes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Parents TT x tt (1); F1 is Tt, all tall (1); Gametes of F1 are T and t (1); F2 has TT, Tt, Tt, tt (1); Ratio is 3 tall : 1 dwarf (1).	1+1+1+1+1
	OR: Allele: alternative form of a gene (1); Homozygous: same alleles (1); Heterozygous: different alleles (1); F1: first filial generation (1); F2: second filial generation (1).	1+1+1+1+1
20	Parents RRYY x rryy (1); F1 is RrYy round yellow (1); F2 gives four phenotypes (1); Ratio 9 : 3 : 3 : 1 (1); New combinations show independent assortment (1).	1+1+1+1+1
	OR: Genes are on chromosomes (1); Each trait is controlled by a pair of genes (1); One gene of each pair comes from each parent (1); Gametes carry one of each pair (1); Zygote has the full pair (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.
- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** sex chromosomes	X and Y.
2	**(b)** heterozygous tall	Tt.
3	**(b)** DNA and protein	Genetic material.
4	**(c)** one X chromosome as a sex chromosome	Each egg has an X.
5	**(d)** 3 : 1	Three dominant to one recessive.
6	**(b)** differences in characters	Variation is the basis of diversity.
7	**(d)** X or Y	Half the sperms have X, half have Y.
8	**(c)** 9 : 3 : 3 : 1	Four types of phenotypes.
9	**(a)** tall	T is dominant over t.
10	**(a)** a girl	XX is female.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A unit of heredity (1); A segment of DNA (1).	1+1
12	Alternative form of a gene (2).	2
13	Differences in characters among individuals (2).	2
14	Male: XY (1); Female: XX (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Chromosomes other than the sex chromosomes (1); 22 pairs (1); 44 autosomes in all (1).	1+1+1
16	Eye colour (1); Hair colour (1); Blood group (1).	1+1+1
17	Inherited traits are in genes (1); Acquired traits are due to environment or use (1); Only inherited traits pass to offspring (1).	1+1+1
	OR: Genotype is genetic make-up, e.g. Tt (1); Phenotype is outward appearance, e.g. tall (1); One phenotype can have different genotypes (1).	1+1+1
18	Mother has two X chromosomes (1); All her eggs have X (1); Only the sperm can bring Y (1).	1+1+1
	OR: Y-carrying sperm gives a boy (1); X-carrying sperm gives a girl (1); Both are equally likely (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Female has XX (1); Male has XY (1); Egg has X (1); Sperm may have X or Y (1); Boy or girl each have 50% chance (1).	1+1+1+1+1
	OR: Parents Tt x Tt (1); Gametes T, t and T, t (1); Offspring TT, Tt, Tt, tt (1); tt is dwarf (1); Dwarf are 25% (1).	1+1+1+1+1
20	Inherited: passed by genes (1); Acquired: gained during life (1); Inherited: eye colour, hair colour (1); Acquired: scar, tan (1); Acquired traits are not inherited (1).	1+1+1+1+1
	OR: 3 : 1 shows dominance and segregation (1); Mendel obtained it in the monohybrid cross (1); 9 : 3 : 3 : 1 shows independent assortment (1); These are obtained in dihybrid crosses (1); They show traits are controlled by factors (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.

- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** half the number of chromosomes	Haploid.
2	**(d)** wrinkled	r is recessive.
3	**(c)** a girl	XX.
4	**(d)** full number of chromosomes	Zygote has 46.
5	**(a)** tall and dwarf	Both are forms of the same character.
6	**(c)** 50%	Equal probabilities.
7	**(d)** No	It is acquired.
8	**(d)** F1	First filial generation.
9	**(b)** Father	Because the sperm may carry X or Y.
10	**(a)** different forms of a gene	Alternate forms.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Scar (1); Tanned skin or muscles (1).	1+1
12	Garden pea (1); Pisum sativum (1).	1+1
13	First filial generation (2).	2
14	Father (1); His sperm carries X or Y (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pure tall TT (1); Pure dwarf tt (1); F1 Tt (1).	1+1+1
16	Through gametes (1); Each parent gives half (1); Zygote has the full number (1).	1+1+1
17	Yellow and green (1); Round and wrinkled (1); Yellow and round are dominant (1).	1+1+1
	OR: Tall 3/4 (1); Dwarf 1/4 (1); Ratio 3 : 1 (1).	1+1+1
18	Each gamete has a different gene combination (1); Fertilisation combines them differently (1); So each child is unique (1).	1+1+1
	OR: Dihybrid cross (1); F2 ratio is 9 : 3 : 3 : 1 (1); Four types of phenotype (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Gametes carry different genes (1); Fertilisation combines parents' genes (1); Copying errors occur in DNA (1); New combinations are formed (1); Variation helps the species survive (1).	1+1+1+1+1
	OR: Gametes have half the number (1); Sperm has 23 (1); Egg has 23 (1); Zygote has 46 (1); So every generation has 46 (1).	1+1+1+1+1
20	Many contrasting traits (1); Short life cycle (1); Self-pollinating and easy to cross artificially (1); Produces many seeds (1); Easy to grow (1).	1+1+1+1+1
	OR: F1 has two alleles (1); Dominant allele masks recessive (1); Alleles separate in gametes (1); In F2 two recessives may combine (1); The recessive trait then appears (1).	1+1+1+1+1

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